

EXP NO: 1	SETTING UP THE ENVIRONMENT AND PREPROCESSING THE DATA
Date:16-7-26	

AIM:

To set up a fully functional machine learning development environment and to perform data preprocessing operations like handling missing values, encoding categorical variables, feature scaling, and splitting datasets.

ALGORITHM:

1. Install Required Libraries:

- Install numpy, pandas, matplotlib, seaborn, and scikit-learn using pip.

2. Import Libraries.

3. Load Dataset:

- Load any dataset (e.g., Netflix or Iris) using pandas.

4. Data Exploration:

- Use `df.info()`, `df.describe()`, `df.isnull().sum()` to understand the data.

5. Handle Missing Values:

- Use `.fillna()` or `.dropna()` depending on the strategy.

6. Encode Categorical Data:

- Use `pd.get_dummies()` or `LabelEncoder`.

7. Feature Scaling:

- Normalize or standardize the numerical features using `StandardScaler` or `MinMaxScaler`.

8. Split Dataset:

- Use `train_test_split()` from `sklearn` to create training and testing sets.

9. Display the Preprocessed Data.

CODE:

```
In [3]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.preprocessing import LabelEncoder, StandardScaler
from sklearn.model_selection import train_test_split
df = pd.read_csv("each_match_records.csv")
print("===== First 5 Rows =====")
print(df.head())
print("\n===== Dataset Information =====")
print(df.info())
print("\n===== Statistical Summary =====")
print(df.describe(include='all'))

print("\n===== Missing Values =====")
print(df.isnull().sum())
numeric_columns = df.select_dtypes(include=np.number).columns
for col in numeric_columns:
    df[col].fillna(df[col].mean(), inplace=True)
categorical_columns = df.select_dtypes(include='object').columns
for col in categorical_columns:
    df[col].fillna(df[col].mode()[0], inplace=True)
print("\nMissing Values After Cleaning")
print(df.isnull().sum())
label_encoder = LabelEncoder()
for col in categorical_columns:
    df[col] = label_encoder.fit_transform(df[col])
print("\nCategorical Columns Encoded Successfully")
scaler = StandardScaler()
df_scaled = pd.DataFrame(
    scaler.fit_transform(df),
    columns=df.columns
)
print("\n===== Scaled Dataset =====")
print(df_scaled.head())
X = df_scaled.iloc[:, :-1]
y = df_scaled.iloc[:, -1]
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42
)
print("\nTraining Features Shape :", X_train.shape)
print("Testing Features Shape  :", X_test.shape)
print("Training Target Shape   :", y_train.shape)
print("Testing Target Shape    :", y_test.shape)
print("\n===== Final Preprocessed Dataset =====")
print(df_scaled.head())
```

OUTPUT:

===== First 5 Rows =====

	season	date	match_number	match_type	\
0	2023	31-03-2023	1	Group	
1	2023	01-04-2023	2	Group	
2	2023	01-04-2023	3	Group	
3	2023	02-04-2023	4	Group	
4	2023	02-04-2023	5	Group	

		venue	location	\
0		Narendra Modi Stadium	Ahmedabad	
1	Punjab Cricket Association IS Bindra Stadium		Chandigarh	
2	Bharat Ratna Shri Atal Bihari Vajpayee Ekana C...		Lucknow	
3	Rajiv Gandhi International Stadium		Hyderabad	
4		M Chinnaswamy Stadium	Bengaluru	

	team1	team2	\
0	Chennai Super Kings	Gujarat Titans	
1	Punjab Kings	Kolkata Knight Riders	
2	Lucknow Super Giants	Delhi Capitals	
3	Rajasthan Royals	Sunrisers Hyderabad	
4	Mumbai Indians	Royal Challengers Bangalore	

		toss_won	toss_decision	umpire1	\
0		Gujarat Titans	field	Nitin Menon	
1		Kolkata Knight Riders	field	BNJ Oxenford	
2		Delhi Capitals	field	AK Chaudhary	
3		Sunrisers Hyderabad	field	KN Ananthapadmanabhan	
4		Royal Challengers Bangalore	field	Nitin Menon	

	umpire2	reserve_umpire	match_referee	\
0	HAS Khalid	A Bengeri	J Srinath	
1	YC Barde	PM Joshi	M Nayyar	
2	NA Patwardhan	M Kuppuraj	DS Manohar	
3	R Pandit	Abhijit Bhattacharya	V Narayan Kutty	
4	Tapan Sharma	A Bengeri	J Srinath	

	winner	winner_runs	winner_wickets	man_of_match
0	Gujarat Titans	NaN	5.0	Rashid Khan
1	Punjab Kings	7.0	NaN	Arshdeep Singh
2	Lucknow Super Giants	50.0	NaN	MA Wood
3	Rajasthan Royals	72.0	NaN	JC Buttler
4	Royal Challengers Bangalore	NaN	8.0	F du Plessis

===== Dataset Information =====

<class 'pandas.core.frame.DataFrame'>

RangeIndex: 74 entries, 0 to 73

Data columns (total 18 columns):

#	Column	Non-Null Count	Dtype
0	season	74 non-null	int64
1	date	74 non-null	object
2	match_number	74 non-null	int64
3	match_type	74 non-null	object
4	venue	74 non-null	object
5	location	74 non-null	object
6	team1	74 non-null	object
7	team2	74 non-null	object
8	toss_won	74 non-null	object
9	toss_decision	74 non-null	object
10	umpire1	74 non-null	object
11	umpire2	74 non-null	object
12	reserve_umpire	74 non-null	object
13	match_referee	74 non-null	object
14	winner	74 non-null	object
15	winner_runs	40 non-null	float64
16	winner_wickets	33 non-null	float64
17	man_of_match	73 non-null	object

dtypes: float64(2), int64(2), object(14)

memory usage: 10.5+ KB

None

===== Statistical Summary =====

	season	date	match_number	match_type	venue \
count	74.0	74	74.000000	74	74
unique	NaN	56	NaN	4	12
top	NaN	14-05-2023	NaN	Group	Narendra Modi Stadium
freq	NaN	2	NaN	70	9
mean	2023.0	NaN	37.500000	NaN	NaN
std	0.0	NaN	21.505813	NaN	NaN
min	2023.0	NaN	1.000000	NaN	NaN
25%	2023.0	NaN	19.250000	NaN	NaN
50%	2023.0	NaN	37.500000	NaN	NaN
75%	2023.0	NaN	55.750000	NaN	NaN
max	2023.0	NaN	74.000000	NaN	NaN

	location	team1	team2	toss_won \
count	74	74	74	74
unique	12	10	10	10
top	Ahmedabad	Chennai Super Kings	Gujarat Titans	Rajasthan Royals
freq	9	10	10	10
mean	NaN	NaN	NaN	NaN
std	NaN	NaN	NaN	NaN
min	NaN	NaN	NaN	NaN
25%	NaN	NaN	NaN	NaN
50%	NaN	NaN	NaN	NaN
75%	NaN	NaN	NaN	NaN
max	NaN	NaN	NaN	NaN

	toss_decision	umpire1	umpire2	reserve_umpire	match_referee	\
count	74	74	74	74	74	
unique	2	15	17	12	13	
top	field	Nitin Menon	VK Sharma	A Bengeri	J Srinath	
freq	53	11	10	9	12	
mean	NaN	NaN	NaN	NaN	NaN	
std	NaN	NaN	NaN	NaN	NaN	
min	NaN	NaN	NaN	NaN	NaN	
25%	NaN	NaN	NaN	NaN	NaN	
50%	NaN	NaN	NaN	NaN	NaN	
75%	NaN	NaN	NaN	NaN	NaN	
max	NaN	NaN	NaN	NaN	NaN	

	winner	winner_runs	winner_wickets	man_of_match
count	74	40.000000	33.000000	73
unique	11	NaN	NaN	51
top	Gujarat Titans	NaN	NaN	Shubman Gill
freq	11	NaN	NaN	4
mean	NaN	30.400000	5.727273	NaN
std	NaN	27.554887	1.908414	NaN
min	NaN	1.000000	1.000000	NaN
25%	NaN	7.750000	5.000000	NaN
50%	NaN	22.000000	6.000000	NaN
75%	NaN	51.250000	7.000000	NaN
max	NaN	112.000000	9.000000	NaN

```

===== Missing Values =====
season          0
date            0
match_number    0
match_type      0
venue           0
location        0
team1           0
team2           0
toss_won        0
toss_decision   0
umpire1         0
umpire2         0
reserve_umpire  0
match_referee   0
winner          0
winner_runs     34
winner_wickets  41
man_of_match     1
dtype: int64

```

Missing Values After Cleaning

```
season      0
date        0
match_number 0
match_type  0
venue       0
location    0
team1       0
team2       0
toss_won    0
toss_decision 0
umpire1     0
umpire2     0
reserve_umpire 0
match_referee 0
winner      0
winner_runs 0
winner_wickets 0
man_of_match 0
dtype: int64
```

Categorical Columns Encoded Successfully

===== Scaled Dataset =====

	season	date	match_number	match_type	venue	location	team1	\
0	0.0	1.705356	-1.708801	0.04398	0.366065	-1.422224	-1.541090	
1	0.0	-1.680400	-1.661984	0.04398	0.663743	-0.886628	0.495350	
2	0.0	-1.680400	-1.615168	0.04398	-1.122330	1.255755	-0.183463	
3	0.0	-1.557281	-1.568351	0.04398	0.961422	0.452361	0.834757	
4	0.0	-1.557281	-1.521535	0.04398	-0.229293	-1.154426	0.155944	

	team2	toss_won	toss_decision	umpire1	umpire2	reserve_umpire	\
0	-0.809958	-0.811496	0.629465	1.226042	-1.159783	-1.712146	
1	-0.448893	-0.479724	0.629465	-0.812769	1.427874	0.897212	
2	-1.171024	-1.143268	0.629465	-1.067621	-0.297231	0.027426	
3	1.717502	1.510907	0.629465	0.206636	0.134045	-1.132289	
4	1.356437	1.179135	0.629465	1.226042	0.565321	-1.712146	

	match_referee	winner	winner_runs	winner_wickets	man_of_match
0	-0.888537	-0.774375	0.000000	-5.795161e-01	0.780214
1	-0.640417	0.808601	-1.169771	-7.077313e-16	-1.478725
2	-1.136656	-0.141184	0.979808	-7.077313e-16	-0.415695
3	1.592660	1.125196	2.079593	-7.077313e-16	-0.615013
4	-0.888537	1.441791	0.000000	1.810988e+00	-1.080089

Training Features Shape : (59, 17)

Testing Features Shape : (15, 17)

Training Target Shape : (59,)

Testing Target Shape : (15,)

===== Final Preprocessed Dataset =====

	season	date	match_number	match_type	venue	location	team1	\
0	0.0	1.705356	-1.708801	0.04398	0.366065	-1.422224	-1.541090	
1	0.0	-1.680400	-1.661984	0.04398	0.663743	-0.886628	0.495350	
2	0.0	-1.680400	-1.615168	0.04398	-1.122330	1.255755	-0.183463	
3	0.0	-1.557281	-1.568351	0.04398	0.961422	0.452361	0.834757	
4	0.0	-1.557281	-1.521535	0.04398	-0.229293	-1.154426	0.155944	

	team2	toss_won	toss_decision	umpire1	umpire2	reserve_umpire	\
0	-0.809958	-0.811496	0.629465	1.226042	-1.159783	-1.712146	
1	-0.448893	-0.479724	0.629465	-0.812769	1.427874	0.897212	
2	-1.171024	-1.143268	0.629465	-1.067621	-0.297231	0.027426	
3	1.717502	1.510907	0.629465	0.206636	0.134045	-1.132289	
4	1.356437	1.179135	0.629465	1.226042	0.565321	-1.712146	

	match_referee	winner	winner_runs	winner_wickets	man_of_match
0	-0.888537	-0.774375	0.000000	-5.795161e-01	0.780214
1	-0.640417	0.808601	-1.169771	-7.077313e-16	-1.478725
2	-1.136656	-0.141184	0.979808	-7.077313e-16	-0.415695
3	1.592660	1.125196	2.079593	-7.077313e-16	-0.615013
4	-0.888537	1.441791	0.000000	1.810988e+00	-1.080089

RESULT:

The Python environment was successfully set up and the dataset was pre-processed by handling missing values, encoding categorical data, performing feature scaling, and splitting the data into training and testing sets. The dataset is now ready for model training and analysis.